

Reflection

I found the feedback to be quite helpful for this project, especially near the end, as I had a few bugs that I'd completely forgotten to fix, and it gave me a lot of useful tips for how to make my database & website better. The biggest changes I made were getting rid of the "Purchases" table, which I had envisioned was for selling guitars to the store that would then be added to the list of guitars to sell, but it was identical to the "Sales" table, and felt redundant and unnecessary. The other big change was adding the lessons table as I didn't have an M:N relationship initially, and I figured that would be the only way that I could reasonably create that relationship between the two tables. Though I would say development of the project was pretty linear and I felt good about it the whole time, even if there are still some things I would like to work on. In all, the feedback was helpful, and I found the project rewarding, and, at times, quite fun.

Seasick Steve's Guitar Emporium

Team Members

- Thomas Munro, Nolan Flores

Overview

- Seasick Steve's Guitar Emporium is struggling to maintain their growth as their store gains more traction. This business buys and sells new or used guitars & basses to musicians with two locations, one in Astoria, OR, and another in Newport, OR. The store also provides guitar, bass, piano, drum, and vocal lessons for students at a rate of \$25 for half an hour, or \$50 for an hour. They've been steadily growing their business over the past few years, and have gotten to

a point where they need to upgrade their backend in order to support new customers and further growth for the business. More customers means a need for better management of inventory for the products being sold, and better ways to keep track of each sale or purchase that the shop makes. The idea is to streamline the flow of commerce and ensure that customers will get an accurate idea of the gear that they're buying.

Link

- <http://classwork.engr.oregonstate.edu:24087/>

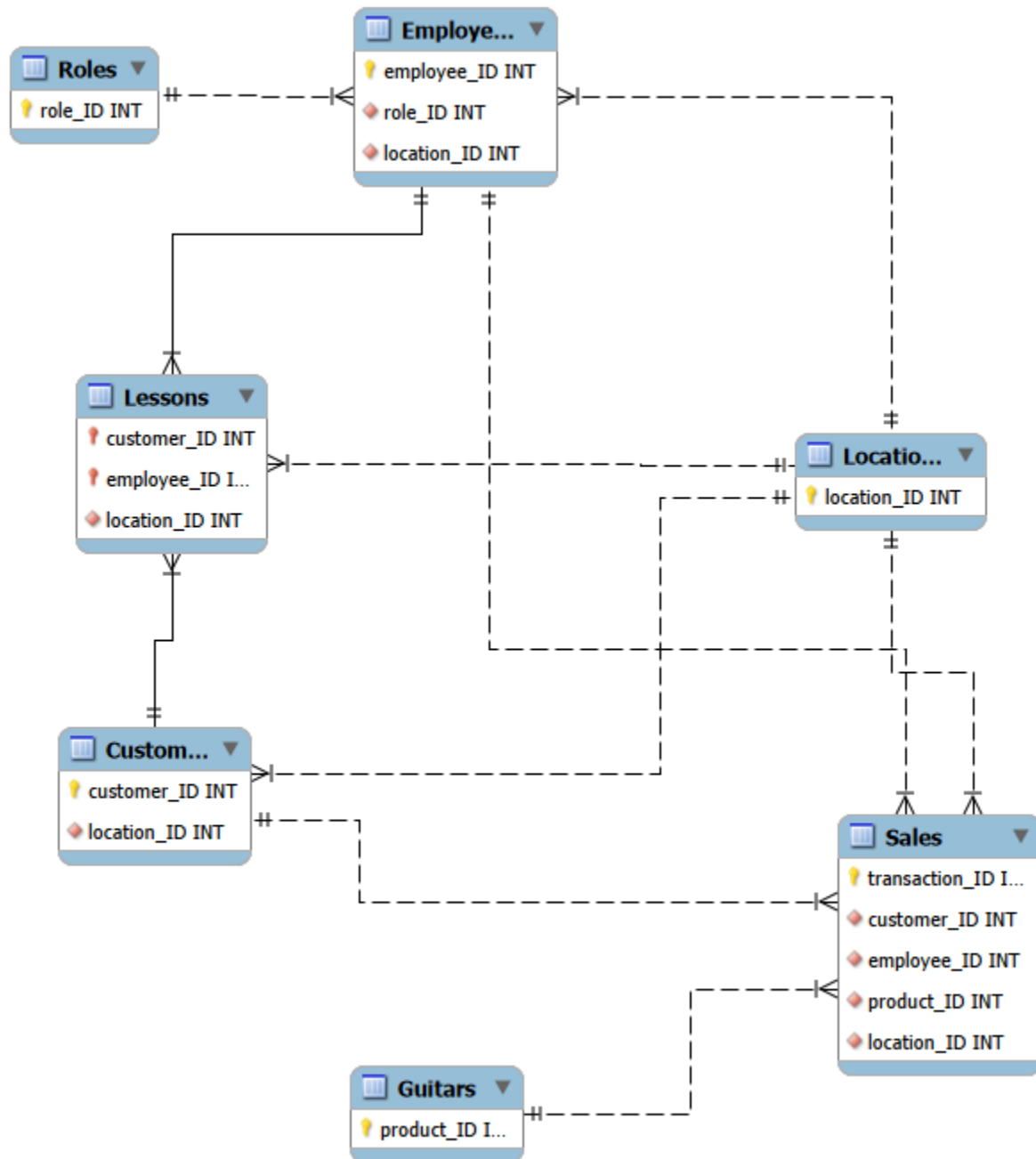
Database Outline

- Sales
 - Tabulates all customer sales for all products, including the guitar that was sold, the customer, the employee who sold the items, and the location where the item was sold.
 - transaction_ID: INT, auto_increment, unique, not NULL, PK
 - customer_ID: INT, not NULL, FK
 - employee_ID: INT, not NULL, FK
 - product_ID: INT, not NULL, FK
 - s_date: DATE, not NULL
 - amount: FLOAT, not NULL, format of 00.00
 - location_ID: INT, not NULL, FK
 - Relationships
 - M:1 relationship from Sales to Customers
 - M:1 relationship from Sales to Employees
 - M:1 relationship with Guitars
 - M:1 with Locations
- Locations
 - Stores all store locations, along with each store's address and phone number.
 - location_ID: INT, not NULL, unique
 - l_address: varchar(45), not NULL
 - phone_number: varchar(45), not NULL
 - Relationships
 - 1:M relationship with Employees

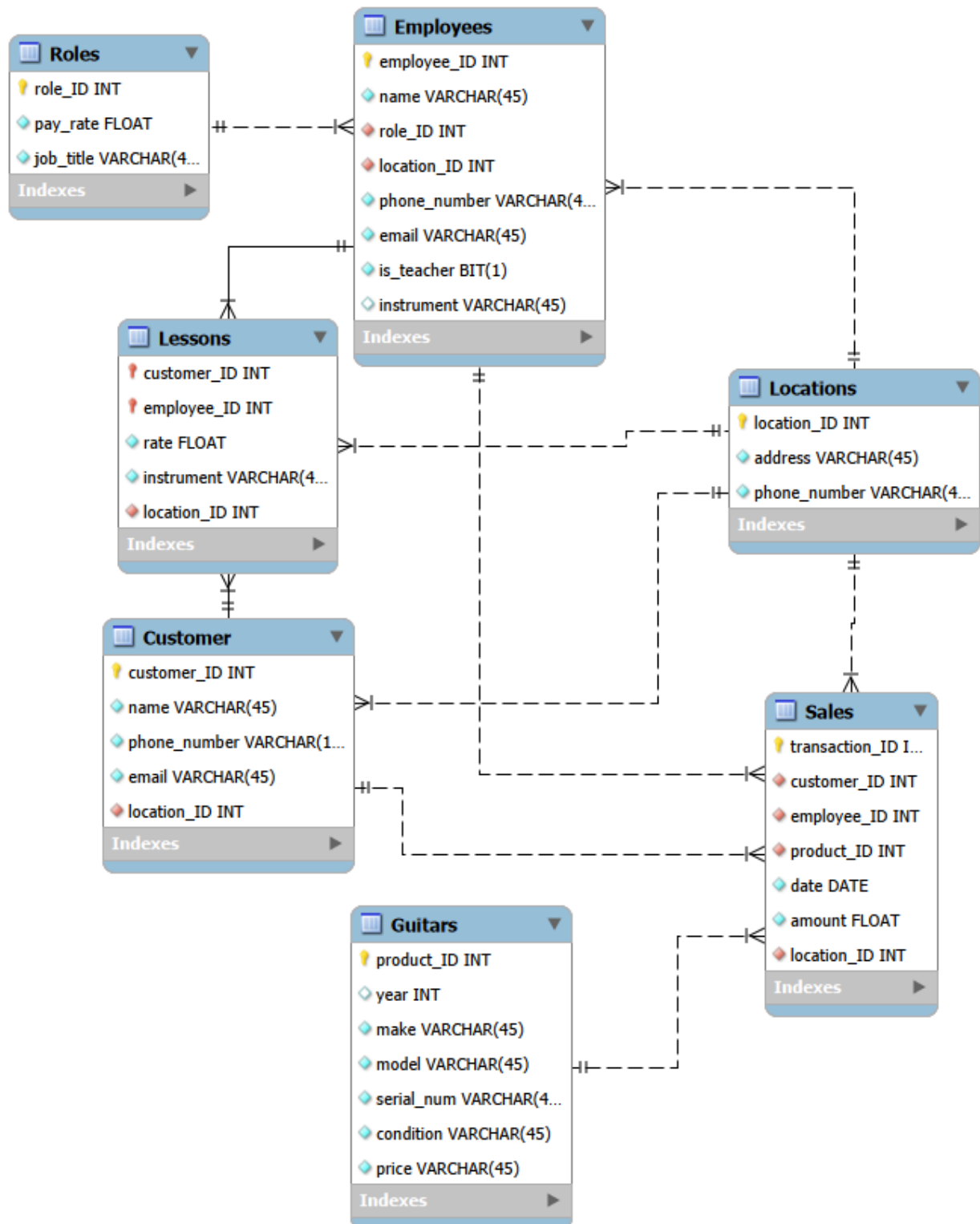
- 1:M relationship with Sales
 - 1:M relationship with Lessons
 - 1:M relationship with Purchases
- Employees
 - This entity provides information for each employee that works at the shop. This is to keep track of their pay, their role, the location they work, and keep track of who makes sales.
 - employee_ID: INT, auto_increment, unique, not NULL, PK
 - e_name: varchar(45), not NULL
 - role_ID: INT, not NULL, FK
 - location_ID: INT, not NULL, FK
 - phone_number: varchar(45), not NULL
 - email: varchar(45), not NULL
 - is_Teacher: BIT(1), not NULL
 - instrument: varchar(45)
 - Relationships
 - 1:M relationship with Sales
 - 1:M relationship with Purchases
 - M:1 relationship with Locations
 - M:N relationship with Customers through Lessons
- Roles
 - This entity keeps track of the different job titles that employees will have, along with their rate of pay.
 - role_ID: INT, not NULL, unique
 - pay_rate: FLOAT, not NULL, format of 00.00
 - job_title: varchar(45)
 - Relationships
 - 1:M relationship with Employees
- Customers
 - This entity keeps track of all customers who do business with the company.
 - customer_ID: INT, auto_increment, unique, not NULL, PK
 - c_name: varchar(45), not NULL
 - phone_number: varchar(16), not NULL
 - email: varchar(45), not NULL
 - Relationships
 - 1:M relationship with Sales

- 1:M relationship with Purchases
 - M:N relationship with Employees through Lessons
- Guitars
 - This entity keeps track of all guitars & basses in the inventory, and stores information about the make, model, year, condition, and serial number (a unique number printed onto a guitar, different from the product number the store gives).
 - product_ID: INT, auto_increment, unique, not NULL, PK
 - year: INT, not NULL
 - brand: varchar(45), not NULL
 - model: varchar(45), not NULL
 - serial_num: varchar(45), not NULL
 - condition: varchar(45), not NULL
 - amount: float, not NULL, format of 00.00
 - Relationships
 - 1:M relationship with Sales
 - 1:M relationship with Purchases
- Lessons
 - This entity keeps track of which customers take lessons with which employee, along with the rate, the instrument being taught, and the location of the lessons. Customers can take lessons with multiple employees, and employees will have multiple customers as their students.
 - customer_ID: INT, not NULL, FK
 - employee_ID: INT, not NULL, FK
 - rate: FLOAT, not NULL, format of 00.00
 - instrument: varchar(45), NOT NULL
 - location_ID: INT, not NULL, FK
 - Relationships
 - 1:M relationship with Customers
 - 1:M relationship with Employees
 - M:1 relationship with Locations

ERD Diagram



Schema Diagram



Tables

Employees								Locations			Roles		
employee_ID	name	role_ID	location_ID	phone_num	email	is_teacher	instruments	location_ID	address	phone_number	role_ID	pay_rate	job_title
1	James Jamerson		1	2 9713292104	email1@gmail.com	0	NULL	1	Newport, OR	9713367324	1	\$16	Sales
2	Jim Johnman		1	1 9713292105	email2@gmail.com	0	NULL	2	Astoria, OR	9713292108	2	\$25	Teacher
3	John Jimman		2	2 9713292106	email3@gmail.com	1	Guitar/Bass				3	\$30	Manager
4	Man Jimjohn		1	1 9713292108	email4@gmail.com	0	NULL						
5	Steve Screeb		3	2 9713367324	email5@gmail.com	0	NULL						
6	Carol Crumb		3	1 9713343532	email6@gmail.com	0	NULL						
Customers													
Customer_ID	name	phone_number	email	is_Student									
1	John Smith	9713282104	email6@gmail.com	1									
2	Smith John	9723487283	email5@gmail.com	1									
3	Cruella Deville	9713282105	email2@gmail.com	1									
Guitars													
product_ID	year	make	model	condition	price								
1	1973	Fender	Precision Bass	Good	\$3,000								
2	2009	Epiphone	Riviera	Great	\$900								
3	1962	Fender	Telecaster	Good	\$5,000								
4	2022	Fender	Player Jazz Bass	Good	\$600								
5	2013	Gibson	Les Paul	Mint	\$3,000								
Sales													
transaction_ID	customer_ID	employee_ID	product_ID	location_ID	date	amount							
1	1	1	2	3	1 3/12/2024	\$3,000							
2		2	2	4	2 4/9/2024	\$900							
3		3	4	5	2 6/13/2024	\$5,000							
Lessons													
customer_ID	employee_ID	rate	instrument	location_ID									
1		3	\$25 Guitar	2									
2		3	\$25 Guitar	2									
3		3	\$25 Bass	2									
Purchases													
transaction_ID	customer_ID	employee_ID	product_ID	location_ID	date	amount							
1		1	2	3	1 3/12/2024	\$3,000							
2		2	2	4	2 4/9/2024	\$900							
3		3	4	5	2 6/13/2024	\$5,000							

Screenshots

DISPLAY Sales

S.S Steve's Guitar Emporium DB

RESET DATABASESalesGuitarsEmployeesLessonsLocationsCustomersRoles

ENTITY PAGE

Sales

Browse & manage sales records.

Browse Sales

SELECT

transaction_ID	customer_ID	employee_ID	product_ID	location_ID	s_date	amount
1	1	2	3	1	Tue Mar 12 2024 00:00:00 GMT-0700 (Pacific Daylight Time)	3000
2	2	2	4	2	Tue Apr 09 2024 00:00:00 GMT-0700 (Pacific Daylight Time)	900
3	3	4	5	2	Thu Jun 13 2024 00:00:00 GMT-0700 (Pacific Daylight Time)	5000
4	3	3	2	1	Sun Nov 17 2024 00:00:00 GMT-0800 (Pacific Standard Time)	2000

Add Sales

INSERT DRAFT

Customer ID

Employee ID

Product ID

Location ID

Date of Sale

Sale Amount

mm / dd / yyyy

mm / dd / yyyy

Update Sales

UPDATE DRAFT

Customer ID

Employee ID

Product ID

Location ID

Date of Sale

Sale Amount

mm / dd / yyyy

mm / dd / yyyy

INSERT Sales

Add Sales

INSERT DRAFT

Customer ID

1

Employee ID

2

Product ID

3

Location ID

2

Date of Sale

06 / 01 / 2026

Sale Amount

2000

Add Sale

DISPLAY Guitars

S.S Steve's Guitar Emporium DB

RESET DATABASESalesGuitarsEmployeesLessonsLocationsCustomersRoles

ENTITY PAGE

Guitars

Browse & manage guitar inventory.

Browse Guitars

SELECT DRAFT

product_ID	g_year	amount	brand	model	g_condition	serial_num
1	1973	3000	Fender	Precision Bass	Good	1001231
2	2009	900	Epiphone	Riviera	Great	2001352
3	1962	5000	Fender	Telecaster	Good	3333333
4	2022	600	Fender	Player Jazz Bass	Good	6666666
5	2013	3000	Gibson	Les Paul	Mint	4242424

Add Guitars

INSERT DRAFT

Year

Price

Brand

Model

Condition

Serial Number

Update Guitars

UPDATE DRAFT

Guitar ID

Year

Price

Brand

Model

Condition

DISPLAY Employees

S.S Steve's Guitar Emporium DB

RESET DATABASESalesGuitarsEmployeesLessonsLocationsCustomersRoles

ENTITY PAGE

Employees

Browse & manage employee records.

Browse Employees

SELECT DRAFT

employee_ID	e_name	role_ID	location_ID	phone_number	email	is_Teacher	Instrument
1	James Jamerson	1	2	971-329-2104	email1@gmail.com	0	
2	Jim Johnman	1	1	971-329-2105	email2@gmail.com	0	
3	John Jimman	2	2	971-329-2165	email3@gmail.com	1	Guitar/Bass
4	Man Jimjohn	1	1	971-329-2175	email@gmail.com	0	
5	Steve Screeb	3	2	971-339-2175	mail@gmail.com	0	
6	Carol Crumb	3	1	971-349-2175	mail2@gmail.com	0	

Add Employees

INSERT DRAFT

Name

Role ID

Location ID

Phone Number

Update Employees

UPDATE DRAFT

Employee ID

Role ID

Name

DISPLAY Lessons

S.S Steve's Guitar Emporium DB

RESET DATABASESalesGuitarsEmployeesLessonsLocationsCustomersRoles

ENTITY PAGE

Lessons

Browse & manage lessons.

Browse Lessons

SELECT DRAFT

customer_ID	employee_ID	location_ID	rate	instrument
1	3	2	25	Guitar
2	3	2	25	Guitar
3	3	2	25	Bass

Add Lessons

INSERT DRAFT

Customer ID

Employee ID

Rate

Instrument

Location ID

Update Lessons

UPDATE DRAFT

Customer ID

Employee ID

Rate

Instrument

Location ID

UPDATE Lessons

Update Lessons

UPDATE DRAFT

Customer ID

1

Employee ID

3

Rate

20

Instrument

Bass

Location ID

1

Update Lesson

DELETE Lessons

Delete Lessons

DELETE DRAFT

Customer ID

1

Employee ID

3

Delete Lessons

DISPLAY Locations

S.S Steve's Guitar Emporium DB

RESET DATABASESalesGuitarsEmployeesLessonsLocationsCustomersRoles

ENTITY PAGE

Locations

Browse & manage store locations.

location_ID	l_address	phone_number
1	Newport, OR	971-233-2314
2	Astoria, OR	971-230-1234
3	Cannon Beach, OR	503-230-3210

Add Locations

INSERT DRAFT

Address

Phone Number

Add Location

Update Locations

UPDATE DRAFT

Location ID

Address

Phone Number

Update Location

Delete Locations

DELETE DRAFT

DISPLAY Customers

S.S Steve's Guitar Emporium DB

RESET DATABASESalesGuitarsEmployeesLessonsLocationsCustomersRoles

ENTITY PAGE

Customers

Browse & manage customer records.

Browse Customers

SELECT DRAFT

customer_ID	c_name	phone_number	email
1	John Smith	971-340-2165	cmail@gmail.com
2	Smith John	972-348-7283	blackmail@gmail.com
3	Cruella Deville	971-328-2104	ihatedogs@gmail.com

Add Customers

INSERT draft

Customer Name

Customer Email

Phone Number

Add Customer

Update Customers

UPDATE draft

Customer ID

Customer Name

Customer Email

Phone Number

Update Customer

DISPLAY Roles

S.S Steve's Guitar Emporium DB

RESET DATABASESalesGuitarsEmployeesLessonsLocationsCustomersRoles

ENTITY PAGE

Roles

Browse & manage employee roles.

Browse Roles

SELECT DRAFT

role_ID	pay_rate	job_title
1	16	Sales
2	25	Teacher
3	30	Manager

Add Roles

INSERT DRAFT

Pay Rate

Job Title

Add Role

Update Roles

UPDATE DRAFT

Role ID

Pay Rate

Job Title

Update Role